Felipe Cordeiro de Sousa

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EDUCATION

University Federal of Ceará Fortaleza, Brasil

Bachelor of Engineering - Computer; **Feb 2024 - Current**

Bachelor of Engineering - Mechanical; **Jan 2021 - Feb 2024**

SKILLS SUMMARY

- **Languages**: Python, C++, SQL, Typescript, Javascript.
- **Frameworks/Libs**: Pytorch, Tensorflow, OpenCV, Scikit-Learn, LangChain, HuggingFace, Pandas/Dask, Numpy, Django, Angular.
- **Tools**: Overleaf, MySQL, PostgreSQL, Excel.
- **Platforms**: AWS, Visual Studio Code, Google Colab, Jupyter Notebook, PyCharm.
- **Soft Skills**: Conflict Resolutions, Adaptability, Excellent communication Critical Thinking, Decision-Making.

WORK EXPERIENCE

Research | Biomedical Data Analytics | LINK **Jan 24 - Current**

- ○ Implementation and experiments of Machine Learning models — SVM, Regression, Decision Tree and Clustering.
- ○ Implemented and experiments with Deep learning architectures — CNNs, RCNNs, Transformers;
- ○ Processing Digital Image using Filter and Morphology Mathematics, Canny algorithms.
- ○ Libraries used — Pytorch, LangChain, HuggingFace, Keras, OpenCV, Scikit-Image, and Scipy.

Data Scientist (Samsung Project - AI4Wellness) | Insight Data Science Lab | LINK **June 24 - Current**

- ○ Creation of end-to-end machine learning pipelines using Airflow.
- ○ AutoML using AutoGluon for regression and classification tasks on multimodal data.
- ○ Serving, monitoring, and deployment of machine learning models using SeldonCore, Prometheus, Grafana, and MLflow.

Generative AI Developer | NUVEN | LINK **April 24 - Aug 24**

- ○ Study and experimentation with Large Language Models (LLMs).
- ○ Use of advanced techniques for improving LLMs, such as fine-tuning (LoRA and QLoRA) and RAG (Retrieve Augmented Generation).

Research AI | NUVEN | LINK **June 23 - April 24**

- ○ Implement and optimize algorithms for detection and segmentation tasks, such as YOLOv8 and SAM.
- ○ Studies and experiments on the combination of YOLOv8 for detection of objects and OCR for character recognition.
- ○ Analytical study on the impact of annotation quality on model training using YOLOv8 for detection tasks.

Computer vision research manager | GPAR | LINK **Feb 22 - June 23**

- ○ Implementation of algorithms in python and C++ for real-time object detection using the JETSON NANO.
- ○ Fusing of the sensor LiDAR and Stereo Camera for the best result of object detection using Matlab and ROS.

PROJECTS

Multimodal Agent-Based Architecture for Parkinson's Diagnosis **Sept 25 - Dec 25**

- ○ Development of a scalable IoMT platform using Docker and FastAPI, orchestrating DataOps and MLOps pipelines for real-time processing of audio, sensor, and neuroimaging data.
- ○ Implementation of a Multi-Agent System utilizing Retrieval-Augmented Generation (RAG) and quantized LLMs (Qwen 2.5) with a Late Meta-Fusion strategy to integrate heterogeneous data sources.
- ○ Utilization of Explainable AI (SHAP) and advanced calibration techniques to validate clinical decisions, achieving a 0.93 ROC-AUC and minimizing false negatives.
- ○ Application of SMOTE and specific signal processing techniques (STFT, Wav2Vec) to handle class imbalance and extract robust features from small datasets

Segmentation Medical Imaging using YOLOv8 **July 23 - May 24**

- ○ Utilization of State-of-the-art YOLO model, the YOLOv8 for segmentation medical imaging
- ○ Study and uses of metrics based in spatial-distances for better segmentation measures.
- ○ Utilization of advanced techniques for class imbalance and working small datasets.
- ○ Optimization of hyper parameters such as optimizer, learning rate, class weight.

Advanced approach for object detection in JETSON Nano **Feb 22 - June 23**

- ○ Utilization of YOLOv5 algorithm for near-real-time and accurate for cone detection around ~40 FPS.
- ○ Data and Sensor fusion from camera stereos, YOLOv5 and sensor LiDAR to recognize precise objects in environments.
- ○ Converting coordinates collected from sensor LiDAR to 3D Point Clouds using Matlab Student version.

Biomedical Website **July 23 - Aug 23**

- The Biomedical Website working with DataOps and MLOps section hosted in Kubernetes Docker consuming AWS Services
- The Application has the goal of to annotate the medical imaging datasets (DICOM and NIFTI files) and able to train and predict images using different models
- Front-end application create in angular and back-end built in Flask

Conversion algorithm: "mask2YOLO" **Sept 23 - Nov 23**

- Specialized algorithm for the DICOM/TIFF image dataset
- Implementation of advanced conversion techniques (from 16-bit to 8-bit images) and resizing using interpolation methods.
- Extracting coordinates of pixels from specialist segmentation in the image and transforming to two types of annotations: binary/multi masks and YOLO annotations
- Possibility of conversion from binary/multi masks to YOLO annotations

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CERTIFICATES

Fundamentals of Deep Learning | CERTIFICATE **Sept 2023**

- Introduction to Deep Learning, training of Neural Network, Convolutional Neural Networks architecture.
- Data augmentation techniques and Deployment, Pre-trained models, and Transformers architecture

Artificial Intelligence and Applications | CERTIFICATE **Dec 2023**

- Learned about Supervised and Unsupervised learning, such as Linear models, Nearest Neighbors, Naive Bayes, Decision Tree, Ensemble models, Neural Networks, and Clustering;

PATENTS

- **QHydro — A solution focused on automated water quality analysis, using Artificial Intelligence and QML techniques.** **Dec 2025**

ARTICLES

- **Multimodal Diagnosis of Parkinson's Disease with an Internet-Based Collaborative Agent Architecture of Medical Language Models | Computers in Biology and Medicine (under review)** **Dec 2025**
- **A Multi-Agent AI Approach for Multimodal Analysis and Personalized Management of Parkinson's Disease | CBIC** **Aug 2025**
- **YOLO-based Ankle Ligament Segmentation from Magnetic Resonance Imaging | CBEB** **June 2024**
- **Automatic Anterior Talofibular Ligament Injuries Classification using Ankle MRI | CBEB** **June 2024**
- **Optimizing of the 3D model of VANT using GANs from sustainable materials using Additive Manufacturing | EUs** **Feb 23**